Dataset Design as Ontology Design: Representational Consequences of QA Segmentation in Personal AI Interaction Data

ANONYMOUS AUTHOR(S)

Personalized AI systems increasingly rely on fine-tuning language models on longitudinal interaction data. The dominant practice converts this data into question–answer (QA) pairs, thereby operationalizing interaction as locally independent query–response mappings rather than temporally structured processes. We empirically test this assumption through a longitudinal case study of a single three-year personal interaction corpus (1,122 sessions, 35,756 role-marked turns), constructing two parallel representations of the same underlying data—a cognitive trajectory graph and QA-derived pair sequences—and comparing their structural properties across six controlled conditions.

We find that trajectory-based representations exhibit significantly different transition dynamics from QA-derived ones (KS $D = 0.298$). These differences persist under semantic reconstruction and when evaluated over the full corpus. An order-shuffled counterfactual further demonstrates that temporal ordering—rather than content alone—accounts for the observed structure (KS $D = 0.101$), ruling out content-level explanations for the distributional divergence.

These results suggest that QA slicing removes explicit supervision for higher-order conditional dependencies by collapsing multi-step sequences into independent pairs. This structural consequence has direct implications for training data design in personal AI settings, where individual patterns are amplified rather than averaged out. We provide a replicable pipeline and synthetic example data to support analogous analyses on other corpora.

Additional Key Words and Phrases: personal AI, longitudinal interaction, dataset representation, QA segmentation, cognitive trajectories, data-centric AI

1 INTRODUCTION

As large language models become embedded in long-term knowledge work, a tension has emerged between two conceptions of AI continuity. Existing systems increasingly incorporate forms of cross-session memory: storing facts, tracking preferences, and retrieving past interactions [6, 11, 12]. These mechanisms address *informational* continuity—the system remembers what was said.

However, informational continuity differs from *cognitive* continuity. When a user engages with an AI system over months or years, they are not simply issuing independent queries. They are refining ideas, revising assumptions, and building understanding through iterative exchange. If this process constitutes a form of cognitive trajectory, then the question is not only what the system retains, but whether the structure of evolving thought can be preserved—and eventually learned from.

Recent work has begun to examine how AI systems reshape human cognitive processes through iterative interaction loops [14]. However, existing research has largely focused on interaction-level effects rather than the underlying representational structure of cognition itself—leaving open the question of whether human thought is better modeled as a trajectory than as a static mapping.

The Representation Problem

The dominant practice for fine-tuning personalized AI systems converts interaction data into question–answer (QA) pairs: a user query followed by a model response. This represents a structural choice that encodes a prior assumption about the nature of reasoning: thought is locally conditioned, with each response as a function of the immediately preceding input.

Existing personalization systems (whether they update user-specific adapters [6], manage long-term memory across context tiers [11], or maintain experience streams [12]) operate downstream of this representational assumption. They optimize *how* to use QA-formatted data, but do not question whether that format adequately represents the cognitive processes it purports to capture.

We argue that **dataset design is ontology design**: the choice of how to segment interaction data encodes implicit assumptions about what thinking *is*. For general-purpose AI trained at scale, these assumptions may average out. For personal AI trained on hundreds of conversations from a single user, every structural decision is amplified.

Hypothesis

We formalize this tension as a distinction between two representations:

$$\text{QA representation:} \quad \textit{question} \longrightarrow \textit{answer} \tag{1}$$

$$\text{Trajectory representation:} \quad \textit{state}_t \longrightarrow \textit{state}_{t+1} \tag{2}$$

Under the QA view, each response is conditioned on the current query alone. Under the trajectory view, each cognitive state is conditioned on the accumulated history of reasoning—a higher-order dependency structure:

$$P(x_{t+1} \mid x_t) \qquad \text{[QA: approximates first-order dependency]} \tag{3}$$

$$P(x_{t+1} \mid x_t, x_{t-1}, \ldots) \qquad \text{[Trajectory: higher-order]} \tag{4}$$

This leads to a testable hypothesis: *if real cognitive trajectories encode higher-order, order-dependent structure, then QA-based representations should exhibit systematic differences in their transition dynamics—even when accounting for content and semantic similarity.*

We do not assume that QA representations are entirely incapable of capturing such structure. Rather, we ask whether they empirically do so, and what is lost when they do not.

Study Design and Scope

We adopt a longitudinal case study design, analyzing a single three-year personal interaction corpus in depth. This design reflects a deliberate methodological tradeoff: where cross-sectional studies offer breadth across users, a longitudinal case study enables the kind of depth—tracking structural properties across thousands of sessions, controlling for individual cognitive style, and observing evolution over years—that aggregate approaches cannot achieve with personal AI data. Case study methodology has an established role in HCI research for developing theoretical frameworks and demonstrating the feasibility of new analytical approaches prior to large-scale replication [3, 17]. Our use of it here is similarly motivated: we aim to establish whether trajectory-based representations exhibit analyzable structure at all, and to develop the methodological apparatus for future comparative work across users and platforms.

The corpus comprises 1,122 sessions and 35,756 role-marked turns, spanning three years (2023–2026). We construct two parallel representations of the same underlying data: a *cognitive trajectory graph* that preserves sequential dependencies between intermediate reasoning states, and QA-derived representations constructed at two scales (a sampled subset $n = 3{,}000$ and the full corpus $n = 15{,}384$) to rule out sampling as a confound.

We analyze structural properties at two levels. First, we characterize graph-level properties (refinement chain length, relation type distribution, and long-chain prevalence), comparing against SQuAD [13] as an external QA baseline. Second, we use Δ cos (an embedding-based metric capturing how semantic similarity changes across consecutive steps) to measure second-order transition dynamics across five controlled conditions: random QA concatenation, semantic A→Q stitching, a fully random baseline, an order-shuffled counterfactual, topic-drift stratification.

Key Findings

Across all conditions, we observe consistent and statistically significant differences between trajectory-based and QA-based representations.

Structural collapse. Cognitive trajectory graphs exhibit a mean refinement chain length of 2.15 (max: 13; 8.6% of chains $\geq$4 steps). QA representations (whether sampled n=3,000 or exhaustive n=15,384) collapse to chain length $\approx$1.1, with zero long chains. This pattern holds across sampling scales, ruling out sampling as an explanation.

Non-recoverability. Transition distributions differ significantly between real trajectories and QA-derived sequences (KS $D = 0.298$). Neither random QA concatenation nor semantic A→Q matching recovers the original structure. QA random concatenation yields substantially higher divergence ($D = 0.298$) than semantic A→Q matching ($D = 0.087$), but even the latter remains significantly different from real trajectories, indicating that local semantic continuity is insufficient to reconstruct higher-order dependencies.

Order dependence. Shuffling the temporal order of real trajectory nodes (while preserving content) produces comparable distributional shifts (KS $D = 0.101$), confirming that observed structure depends on ordering rather than content alone.

Robustness. These differences persist in low- and medium-drift strata (KS $D = 0.755/0.381$, both $p < 0.001$); the high-drift stratum is inconclusive because it contains only 19 real transitions ($D = 0.180$, $p = 0.614$). Three distributional metrics converge: KS distance, Wasserstein distance (= 0.188), and KL divergence (= 1.074).

Contributions

This paper makes four contributions.

(1) **A longitudinal case study methodology for analyzing personal AI training data structure.** We demonstrate that a single three-year personal interaction corpus contains analyzable structural properties (refinement chain length, transition dynamics, temporal ordering effects) that are inaccessible to cross-sectional or aggregate approaches. We provide a replicable pipeline enabling researchers to conduct analogous analyses on their own corpora locally, without sharing raw conversation content.

(2) **Cognitive trajectory graphs as a generalizable representational framework.** We introduce a graph schema and construction algorithm applicable to any longitudinal interaction corpus, independent of the specific user or platform, and characterize its structural properties against QA-derived representations at multiple scales.

(3) **Empirical demonstration that QA slicing collapses higher-order sequential dependencies.** Through six controlled conditions and counterfactual constructions on a single three-year corpus (including an order-shuffled counterfactual that isolates temporal ordering as a structural factor independent of content), we show

that this collapse is a structural consequence of the QA format rather than a sampling artifact. These findings motivate multi-user replication as the primary direction for future work.

(4) **"Dataset design is ontology design" as a theoretical framing for personal AI data practice.** We argue that representational choices in training data construction encode implicit assumptions about the nature of cognition, and that in personal AI settings—where data volume is limited and individual patterns are amplified—these choices deserve the same scrutiny as model architecture decisions. We introduce the *consensus agent* as a distinct artifact from a personal cognitive mirror, with different data architecture requirements and evaluation criteria.

Our findings are scoped to structural properties of data representations derived from a single-user longitudinal corpus. Whether these differences produce downstream behavioral gains requires a separate, matched model evaluation and is outside this paper's scope.

2 RELATED WORK

Our work spans four areas of the research stack: model adaptation for personalization, memory architectures for agent continuity, longitudinal human–AI interaction, and training data representation. We review each, focusing on what each layer assumes about interaction data and what it leaves unexamined.

2.1 Personalized Language Models

Personalization of dialogue systems has a long history in HCI and NLP. Early work introduced persona-consistent dialogue, demonstrating that models conditioned on explicit user profiles produce stylistically coherent responses [18]. Subsequent work scaled this approach to larger models and richer user representations [15].

More recently, the challenge has shifted to *continual* personalization: adapting to users whose preferences evolve over time. SPRInG [6] addresses this by maintaining user-specific LoRA adapters [4] and selectively updating them on high-novelty interactions, enabling adaptation to preference drift without catastrophic forgetting. Prefix-tuning [8] and related parameter-efficient methods provide the technical substrate for per-user adaptation.

While these approaches differ in mechanism, they often rely on interaction data representations derived from query–response pairs or sequential dialogue logs, implicitly treating each exchange as conditionally independent given the current query (i.e., each response is a function of the immediately preceding input, without modeling accumulated reasoning history). The question of whether this representational assumption is suitable for capturing personal cognitive processes has not been examined.

2.2 Memory Architectures for AI Agents

A parallel line of work addresses personalization through memory rather than fine-tuning. Generative Agents [12] demonstrated that agents equipped with structured memory—experience storage, reflection, and planning—exhibit coherent long-term behavior, establishing that memory organization is central to sustained behavioral coherence. Other approaches manage memory through OS-inspired paging between an in-context working set and external storage [11].

These approaches primarily provide *informational* continuity, exposing past interactions at inference time without altering the model's underlying inductive biases. Recent work has noted that many graph-based memory architectures rely on entity–relation triples or coarse text chunks as basic units, which may fragment or coarsen the original discourse [2, 19]. This observation highlights that the choice of representation unit itself has structural consequences.

Our work isolates this issue at the training data level: we evaluate whether QA slicing, the most widely used practice, preserves the explicit representation of transition structure in longitudinal interaction data, or removes it from the training signal.

2.3 Longitudinal Human–AI Interaction

HCI research has increasingly recognized that single-session studies cannot capture how human–AI relationships evolve over time. Karapanos et al. [5] identified the need for longitudinal methods in HCI research, noting that single-session evaluations miss temporal dynamics of technology adoption and appropriation. More recent work has highlighted that sustained use of AI systems may reveal evolving patterns (including shifts in mental models, task delegation, and perceived capabilities) that single-session studies cannot capture [9].

Research on the cognitive effects of AI use has begun to characterize how extended interaction reshapes human thought. Kittur et al. [14] document how generative AI alters knowledge work through iterative refinement loops, meta-decision support, and schema induction. They characterize cognition as a dynamic, multi-step process unfolding through iterative state transitions. Related work has examined cognitive load and engagement effects of sustained AI assistance [7].

These studies examine how AI affects human cognition. Our work takes the inverse perspective: how the structural representation of human cognitive processes in interaction logs affects the training of personalized AI.

2.4 Training Data Representation

The dominant paradigm for fine-tuning language models on dialogue data is instruction tuning on query–response pairs [10]. Chain-of-thought prompting [16] demonstrated that preserving intermediate reasoning steps—rather than only final answers—improves performance on complex tasks, establishing that the structure of reasoning, not just its conclusions, carries informational value. Our findings complement this line of work by demonstrating that preserving intermediate reasoning steps is necessary but not sufficient: if the temporal ordering of those steps is destroyed, higher-order trajectory structure collapses regardless of whether individual steps are retained. Despite this, the implications for training data design in personal AI settings have not been examined. Prior work has largely treated data representation as an implementation choice, with limited attention to how segmentation decisions affect the preservation of cognitive structure across turns.

The Gap This Work Addresses

Prior work across HCI and conversational AI has established that single-session evaluations fail to capture the temporal dynamics of human–AI interaction [5], leading to a shift toward longitudinal studies and large-scale interaction data. However, this line of work has primarily focused on behavioral and system-level phenomena, without formalizing the structural dependencies within interaction trajectories themselves.

At the same time, dominant practices in training data construction implicitly assume that interaction can be factorized into independent query–response pairs, treating each step as locally conditioned on the immediately preceding input.

What remains unexamined is whether this representational assumption preserves the higher-order dependencies that may emerge in longitudinal interaction. If such dependencies exist, QA-based segmentation may systematically distort the transition structure of interaction data.

This gap motivates our investigation. We provide the first empirical characterization of whether QA-based representations preserve or distort the transition dynamics of longitudinal personal interaction data.

3 METHOD

We construct two parallel representations of the same longitudinal interaction corpus and compare their structural properties. Section 3.1 describes the cognitive chunking procedure; Section 3.2 defines the resulting graph representation.

3.1 Cognitive Chunking

Raw conversation logs consist of interleaved user and assistant turns. We process them into a cognitive graph $G = (V, E)$ through the three-stage pipeline described in Algorithms 1–3.

Design principles. The chunking algorithm is rule-based rather than learned, a deliberate design choice motivated by four constraints.

Reproducibility. Rule-based segmentation produces identical outputs across runs given the same input, enabling exact replication without model checkpoints or stochastic sampling.

Determinism. Fixed lexical rules avoid the variability introduced by LLM-based segmentation, where prompt sensitivity and version drift can alter chunk boundaries unpredictably.

Interpretability. Each segmentation decision is traceable to an explicit rule (e.g., keyword overlap $< 30\%$ triggers topic shift), facilitating error analysis and allowing researchers to understand why a specific boundary was placed.

Ease of replication. The algorithm requires no fine-tuned models, no API keys, and no proprietary services. Any researcher with the code can apply it to their own corpus locally, addressing the privacy constraints inherent to personal AI data.

The tradeoff is that rule-based detection may miss cognitive events without explicit linguistic markers. We address this limitation in Section 6; importantly, our structural findings (Section 4) are measured on the resulting graphs and are therefore independent of whether future work adopts different segmentation strategies.

Design rationale. Three design choices deserve explicit motivation.

Same-role merging (Algorithm 1, line 6) prevents the graph from encoding unnatural assistant→assistant sequences as reasoning edges, and avoids identity collapse during fine-tuning.

Priority-ordered relation detection (Algorithm 3) ensures that the most semantically meaningful relation type takes precedence when multiple cognitive events co-occur in a single transition.

Iteration-final edges capture the mapping from an initial reasoning state to its converged conclusion, enabling training on the complete arc of a correction chain rather than only its intermediate steps. Only chains of length ≥ 3 (i.e., depth ≥ 2) generate `iteration_final` edges, filtering trivial two-step revisions.

Limitation of current implementation. Cognitive event detection is lexical and rule-based, relying on surface markers to identify transition boundaries. Transitions that occur without explicit linguistic signals—a user implicitly reframing a problem without using contrastive vocabulary—will not trigger a node boundary. Incorporating embedding-based semantic distance ($\Delta \cos$) as a segmentation criterion is a natural extension; we leave this for future work. Importantly, the main findings reported in Section 4 are measured on the resulting graph structure and transition distributions, not derived from the segmentation rules, and are therefore independent of this limitation.

Algorithm 1 Cognitive Node Chunking and Graph Construction

Require: Conversation turns $\mathcal{T} = \{t_1, \ldots, t_n\}$, where $t_i = (\text{role}_i, \text{content}_i)$
Ensure: Cognitive graph $G = (V, E)$
$V \leftarrow \emptyset,\ E \leftarrow \emptyset$
for each turn $t_i \in \mathcal{T}$ **do**
  $S_i \leftarrow$ split t_i.content into sentences
  $V_i \leftarrow \textsc{MergeSentences}(S_i,\ t_i.\text{role})$
  $V \leftarrow V \cup V_i$
end for
$V \leftarrow$ merge consecutive nodes with same role ▷ Prevent consecutive same-role edges
for each consecutive pair $(v_i,\ v_{i+1}) \in V$ **do**
  $r \leftarrow \textsc{DetectRelation}(v_i,\ v_{i+1})$
  $E \leftarrow E \cup \{(v_i,\ v_{i+1},\ r)\}$
end for
$C \leftarrow$ find chains where edge type $\in$ {`refines`, `contrasts`}
for each chain $c = [v_{\text{root}}, \ldots, v_{\text{final}}] \in C$ with $|c| \geq 3$ **do**
  $E \leftarrow E \cup \{(v_{\text{root}},\ v_{\text{final}},\ \texttt{iteration\_final})\}$
end for
return $G = (V, E)$

Algorithm 2 Sentence Merging with Cognitive Event Detection

Require: Sentences $S = \{s_1, \ldots, s_m\}$, role r, minimum node length $\theta_{\min} = 30$ chars
Ensure: Node set V
$V \leftarrow \emptyset,\ v_{\text{curr}} \leftarrow s_1$
for $i = 2$ **to** m **do**
  event $\leftarrow$ `false`
  if $\textsc{TopicShift}(s_i,\ v_{\text{curr}})$ **then** event $\leftarrow$ `true` ▷ Keyword overlap < 30%
  else if $\textsc{ReasoningStep}(s_i)$ **then** event $\leftarrow$ `true` ▷ *so, therefore, thus*
  else if $\textsc{Correction}(s_i)$ **then** event $\leftarrow$ `true` ▷ *but, however, actually*
  else if $\textsc{NewIdea}(s_i)$ **then** event $\leftarrow$ `true` ▷ *what if, maybe*
  else if $\textsc{PerspectiveShift}(s_i)$ **then** event $\leftarrow$ `true` ▷ *from another angle*
  end if
  if event = `true` **and** $|v_{\text{curr}}| \geq \theta_{\min}$ **then**
    $V \leftarrow V \cup \{(v_{\text{curr}},\ r)\}$
    $v_{\text{curr}} \leftarrow s_i$
  else
    $v_{\text{curr}} \leftarrow v_{\text{curr}} \oplus s_i$ ▷ Concatenate
  end if
end for
$V \leftarrow V \cup \{(v_{\text{curr}},\ r)\}$
return V

3.2 Graph Representation

Each conversation is represented as a directed graph $G = (V, E)$.

Nodes. Each node $v \in V$ encodes a cognitive state:

$$v = \{\text{content}, \text{role}, \text{timestamp}, \text{length}\}.$$

Algorithm 3 Relation Type Detection (priority order)

Require: Nodes v_i, v_{i+1}
Ensure: Relation type r

if Correction(v_{i+1}.content) **then**
  return `refines` ▷ Corrects or improves previous thought
else if PerspectiveShift(v_{i+1}.content) **then**
  return `contrasts` ▷ Alternative viewpoint
else if ReasoningStep(v_{i+1}.content) **then**
  return `derives` ▷ Logical inference from prior state
else if v_i.role $\neq$ v_{i+1}.role **then**
  return `responds` ▷ Role transition
else
  return `follows` ▷ Default sequential continuation
end if

Timestamps are preserved from the original conversation records (no missing timestamps observed across all 13,312 nodes in the corpus).

Edges. Each directed edge $e \in E$ carries a relation type from the vocabulary:

{`follows`, `derives`, `refines`, `contrasts`, `responds`, `iteration_final`}.

`iteration_final` edges are additive: they provide a direct shortcut from chain start to chain end without replacing the intermediate edges, enabling training on both step-by-step refinement and its converged outcome.

Extensibility. The graph schema includes a `tags` field reserved for additional relation types (e.g., analogy, hypothesis, example) that may emerge from future unsupervised discovery over larger datasets.

4 EXPERIMENTS

We evaluate the cognitive trajectory graph representation against QA-derived representations through structural analysis of the data itself (graph properties in Section 4.1, trajectory dynamics via $\Delta \cos$ in Section 4.2, and control experiments in Section 4.3).

Figure 1 summarizes the experimental pipeline.

Dataset. The corpus consists of 1,122 personal conversation sessions (35,756 role-marked turns; average 31.9 turns per session, 632 characters per turn) collected over three years (2023–2026). Three representations are constructed from this corpus: **Personal Cognitive** (1,122 trajectory graphs, one per deduplicated session), **Personal QA Sampled** (3,000 QA graphs), and **Personal QA Full** (15,384 QA graphs, covering the entire corpus). We additionally use **SQuAD** [13] (3,000 QA graphs) as an external baseline. Embeddings are computed using `bge-m3` [1]. Session count and graph count are equal (1,122) because each session yields exactly one trajectory graph. Sentence-level segmentation is applied within each turn, but consecutive segments sharing the same role are subsequently merged. A node therefore approximates one conversational turn—or a run of same-role turns—rather than a sub-turn cognitive unit. The ratio of 13,312 nodes to 35,756 role-marked turns (≈ 0.37) reflects this merging.

4.1 Structural Analysis

We compare graph-level structural properties across all four representations (Table 1).

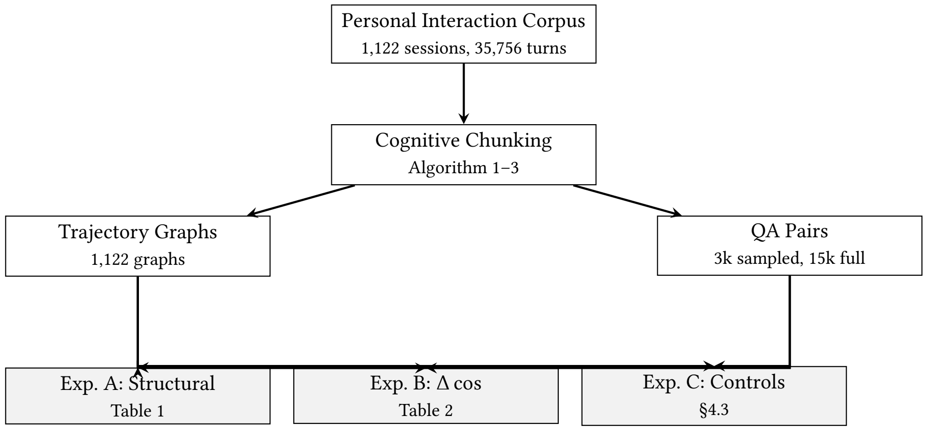

Fig. 1. Experimental pipeline. A single corpus is processed into two parallel representations (trajectory graphs and QA pairs), which are compared across three experiment families: structural properties (A), transition dynamics (B), and robustness controls (C).

Table 1. Structural comparison across dataset representations. Personal QA results are stable across sampled and full-corpus conditions, ruling out sampling as a confound.

Metric	**Cognitive**	**QA (3k)**	**QA (full)**	**SQuAD**
Graphs	1,122	3,000	15,384	3,000
Avg chain length	**2.15**	1.12	1.11	1.08
Max chain length	**13**	2	2	2
Long chain (≥4)	**8.6%**	0.0%	0.0%	0.0%
derives + refines	56.6%	67.0%	66.0%	36.2%
Relation entropy	1.735	1.582	1.609	1.218
Avg input length	365	363	380	127

Refinement chain collapse. The cognitive representation exhibits substantially richer chain structure: mean length 2.15, maximum 13, with 8.6% of chains containing four or more steps. Both QA conditions—sampled and full-corpus—collapse to chain length ≈1.1 with zero long chains and a maximum of two steps. This result holds regardless of sampling scale, directly ruling out sampling bias as an explanation.

The derives+refines paradox. The QA representations show higher *derives+refines* edge proportions (66.0–67.0%) than the cognitive representation (56.6%). This apparent reversal reflects a compression effect: QA segmentation forces multi-turn reasoning into single nodes, concentrating within-pair semantic relations. Chain structure—which requires *cross*-pair dependencies—is eliminated entirely. The two metrics together characterize what QA slicing loses: not semantic density within pairs, but the connective structure across them.

4.2 Trajectory Dynamics

To compare transition dynamics, we compute $\Delta\cos$, a second-order metric capturing how semantic similarity changes across consecutive steps:

$$\Delta\cos(t) = \cos(x_t, x_{t+1}) - \cos(x_{t-1}, x_t) \tag{5}$$

where $\cos(\cdot,\cdot)$ denotes cosine similarity between `bge-m3` embeddings [1]. $\Delta\cos$ captures trajectory *curvature*: positive values indicate increasing semantic coherence (continuation), negative values indicate a direction shift.

We apply this metric to six sequence conditions (Table 2), comparing the $\Delta\cos$ distribution of real cognitive trajectories (Group A) against five alternatives using two-sample KS tests. We focus on KS statistic D as the primary effect-size measure, as p-values saturate under large sample sizes.

Structural divergence (A vs B). Random QA concatenation differs substantially from real trajectories ($D = 0.298$, $p < 0.001$). The ratio of $\Delta\cos$ variances between QA random concatenation and real trajectories (3.82×) indicates that QA sequences exhibit far greater distributional spread. Notably, this elevated variance reflects not richer dynamics but structural discontinuity: the fully random baseline, by contrast, exhibits the narrowest distribution ($\sigma = 0.100$), as independent random transitions produce near-zero second-order changes in high-dimensional embedding space. The high absolute cosine similarity observed in the fully random baseline ($\mu = 0.922$) reflects the narrow semantic range of short text chunks in this embedding space; all structural comparisons rely on the *change* in similarity ($\Delta\cos$) rather than absolute values, controlling for this baseline effect. Real trajectories occupy an intermediate position ($\sigma = 0.183$), with structured semantic acceleration and deceleration producing moderate distributional spread. Cohen's $d = 0.001$ reflects near-identical means across conditions—all $\Delta\cos$ distributions are centered near zero by construction—and should not be interpreted as indicating no effect. The distributional difference captured by KS $D = 0.298$ is a shape difference, not a mean difference: QA sequences and real trajectories differ in variance and modality, not in central tendency.

Semantic reconstruction fails (A vs C). Even when QA pairs are stitched using answer-to-question semantic matching—a method designed to preserve reasoning carryover—the resulting sequences remain significantly different from real trajectories ($D = 0.087$, $p < 0.001$). Local semantic continuity is insufficient to reconstruct higher-order trajectory structure, suggesting that the failure of QA representations is not a consequence of naive concatenation but a structural property of the QA format itself—one that persists even under semantically optimized reconstruction.

Order dependence (A vs F). Shuffling the node order of real trajectories—preserving all content but destroying temporal structure—produces a distributional shift comparable to semantic reconstruction ($D = 0.101$, $p < 0.001$). This experiment functions as a structural intervention on temporal ordering: by permuting node sequence while preserving all segmentation decisions and content, it isolates ordering as an independent determinant of trajectory structure—one that cannot be attributed to differences in vocabulary, topic, or semantic content between groups. When identical content is randomly permuted, higher-order transition dynamics collapse despite preserved semantic distributions, ruling out content-level explanations entirely.

First-order continuity without second-order structure. As a structural baseline, we compute first-order semantic similarity (cosine distance) within QA pairs across all three QA conditions. Personal QA (full corpus) exhibits an average semantic gap of 0.269 (similarity = 0.731); the sampled condition yields 0.277 (similarity = 0.723), confirming sampling stability (difference < 3%). Critically, second-order trajectory dynamics ($\Delta\cos$) are undefined for QA-derived sequences: all QA graphs consist of two-node structures, for which no consecutive triple (x_{t-1}, x_t, x_{t+1}) exists. QA representations preserve local semantic continuity while lacking the multi-step sequential structure required to instantiate trajectory dynamics. This represents a structural absence rather than a performance deficit: the capacity to encode second-order dynamics cannot emerge from a two-node structure by construction. First-order continuity (high

Table 2. $\Delta\cos$ comparison across six sequence conditions. KS D measures distributional distance from real trajectories (A).

Group	Condition	n	KS D vs A
A	Real cognitive trajectories	11,417	–
B	QA random concatenation	2,960	0.298
C	QA semantic A→Q stitching	2,960	0.087
D	Fully random baseline	4,404	0.168
E	SQuAD external baseline	5,748	0.307
F	Shuffled real (order removed)	11,417	0.101

within-pair similarity) and second-order structure (trajectory curvature via $\Delta\cos$) are orthogonal properties; QA slicing preserves the former while making the latter undefined.

Distributional structure. We examine the full $\Delta\cos$ distribution across transitions rather than relying only on threshold-derived summaries. The six conditions form a stable variance gradient: fully random sequences are narrowest ($\sigma = 0.100$), followed by shuffled real trajectories (0.142), real trajectories (0.183), semantic QA stitching (0.205), random QA concatenation (0.358), and SQuAD (0.360). With a fixed reorientation threshold ($\Delta\cos < -0.1$), real trajectories shift on 29.0% of transitions, compared with 47.7% for random QA concatenation, 37.1% for semantic stitching, 14.2% for the fully random baseline, and 20.4% after shuffling real trajectories. The higher QA rate reflects alternating within-pair proximity and between-pair discontinuity, not richer trajectory structure.

Native QA sequences. This distributional analysis applies to QA pseudo-trajectories constructed by concatenation. Native QA-derived sequences—the direct output of QA slicing—consist entirely of two-node graphs, for which no consecutive triple (x_{t-1}, x_t, x_{t+1}) exists. Second-order transition dynamics ($\Delta\cos$) are therefore structurally undefined for native QA, not merely absent as a performance outcome.

4.3 Control Experiments

We conduct three additional controls to verify that observed differences are not artifacts of confounding factors.

Topic drift stratification. We stratify transitions by baseline semantic similarity ($\cos(x_{t-1}, x_t)$) into three drift regimes—low (> 0.7), medium (0.3–0.7), and high (< 0.3)—and apply KS tests within each stratum. Real trajectories and QA sequences differ significantly in the low- and medium-drift regimes ($D = 0.755/0.381$; both $p < 0.001$). The high-drift stratum is small (19 real transitions) and inconclusive ($D = 0.180$, $p = 0.614$). We therefore do not claim robustness under extreme topic shifts; the supported result is that ordinary low- and medium-drift differences are not explained by endpoint similarity.

Sampling invariance. The structural analysis is replicated with the full QA corpus (15,384 graphs) versus the sampled condition (3,000 graphs). Average chain length is stable (1.11 vs 1.12) and long-chain prevalence remains 0.0% in both conditions, ruling out sampling as a confound in the structural results.

Distributional robustness. Beyond KS distance, we report two additional distributional metrics: Wasserstein (Earth Mover) distance (= 0.188) and KL divergence (= 1.074). All three metrics converge in direction and magnitude, providing convergent evidence that the observed differences are not an artifact of any single statistical measure.

5 DISCUSSION

Our findings demonstrate that QA slicing does not merely lose information—it fundamentally alters the statistical structure of cognitive trajectories by collapsing history-dependent state transitions into locally conditioned responses. We discuss five implications.

5.1 Dataset Design Is Ontology Design

The choice of how to segment interaction data is not a neutral preprocessing decision—it encodes an implicit assumption about what thinking *is*.

Our results operationalize this claim: the same underlying interaction corpus, segmented differently, yields representations with measurably different statistical properties.

For general-purpose AI trained at scale, structural assumptions in training data may average out across billions of examples. For personal AI trained on hundreds of conversations from a single user, every representational decision is amplified. The implication is not that QA segmentation is wrong in general, but that its suitability for personal AI cannot be assumed—it must be empirically evaluated.

This observation connects to a broader principle in data-centric AI: the structure imposed on training data determines what patterns a model can learn, independent of model architecture or optimization procedure. In personal AI settings, where the goal is to capture the cognitive style of a specific individual, the ontological commitments embedded in data formatting deserve the same scrutiny as model design choices.

5.2 The "Consensus Agent" and Its Implications

The source corpus is dyadic: user turns redirect the conversation and assistant turns develop, revise, or consolidate responses. The graphs therefore describe structure that emerged through sustained interaction; they do not isolate either participant's private cognition. We use *consensus agent* as a design concept for a system intended to model this co-evolved collaborative history, rather than as a claim about a model evaluated in this paper.

Trajectory representation may be relevant to such an artifact because it retains revision paths that QA slicing removes. This is a statement about the supervision available in the data, not evidence that a fine-tuned model will use that supervision correctly. Establishing the latter requires a separate, leakage-free behavioral comparison.

A consensus agent and a personal cognitive mirror are different artifacts with different use cases.

Table 3. Consensus agent vs. personal cognitive mirror.

Artifact	Primary data	Target behavior
Consensus agent	Dyadic trajectories	Collaborative history
Cognitive mirror	User-side evidence	Individual reasoning

The former targets a long-term collaborative partner shaped by joint dialogue history; the latter requires a data architecture and evaluation protocol that treat user-side contributions as the primary evidence. The present structural study establishes neither artifact; it clarifies that they require different representational commitments.

5.3 Memory Injection vs. Weight Internalization

Our work is motivated by a distinction between two mechanisms of AI personalization: injecting information into context at inference time, and modifying model weights through fine-tuning.

Retrieval-augmented and memory-injection approaches [11, 12] operate at the inference layer: the model is *told* about a user's history at each interaction. Fine-tuning operates at the weight layer: the model's default behavior is shaped by exposure to a user's interaction history.

We hypothesize that these mechanisms differ qualitatively, not just quantitatively. Context injection produces informational continuity: the model knows what happened. Weight modification may produce something closer to cognitive continuity: the model's default reasoning tendencies are shifted. Our results show that representation structure constrains what sequential information is available to a learning system, prior to any choice of storage mechanism.

The analogy is the difference between being *informed* of past experiences and having *lived* through them.

Long context vs. trajectory structure. An immediate question is whether long-context models obviate the need for structured representations. We argue they address different problems. Long context solves an *access* problem: how to expose more interaction history within a single forward pass. Trajectory representation addresses a *training signal* problem: whether the structure of multi-step reasoning is explicitly represented in the data a model is fine-tuned on. A long-context model fine-tuned on QA pairs still receives no explicit supervision for the sequential dependencies that trajectory graphs encode. Trajectory data can expose those dependencies, but whether a model learns to use them is an empirical question. The mechanisms are therefore complementary rather than substitutive.

QA-formatted data exposes locally conditioned transitions. Trajectory data preserves a higher-order signal that a downstream learning method could exploit. The present study establishes the difference in that available signal; it does not claim that preservation alone produces cognitive continuity or better model behavior.

5.4 Structural Difference as a Precondition

Our structural experiments establish that trajectory-based and QA-based representations differ in what they retain. This difference is a precondition for a downstream representation comparison, not its outcome: preserving structure is necessary for learning from it but is not sufficient to guarantee that a model will exploit it.

Standard example-level fine-tuning commonly shuffles samples and optimizes a per-example objective. Such a procedure may ignore the longer-range dependencies preserved in a trajectory graph. A stronger behavioral test therefore requires leakage-free QA and trajectory baselines matched on source sessions, model, budget, and temporal split, plus an order-aware objective or batching strategy where appropriate. We leave that behavioral question to separate work.

5.5 Design Implications

Our findings carry three implications for researchers and practitioners building personal AI systems.

Representation format is a design decision, not a default. The choice to segment interaction data into QA pairs is rarely made explicitly—it is inherited from general-purpose fine-tuning practice. Our results demonstrate that this default carries measurable structural consequences: the same underlying corpus, segmented differently, yields representations with significantly different transition dynamics. Researchers and designers building personal AI systems should treat data format as a first-class design decision, evaluated against explicit criteria for what cognitive structure the format is intended to preserve.

Temporal ordering should be treated as a feature, not metadata. The order-shuffled counterfactual (Group F, KS $D = 0.101$) demonstrates that the structural properties of personal interaction data depend on temporal ordering independently of content. Interaction logs should preserve original conversation timestamps and sequential context as primary structural signals—not merely as provenance metadata—and pipeline designs should treat order-scrambling as a measurable form of information loss.

Distinguish the artifact before choosing the architecture. The consensus agent and the personal cognitive mirror are different artifacts requiring different data architectures and evaluation criteria. A system intended to approximate a long-term collaborative partner would draw on dyadic trajectories structured as graphs. A system intended to capture a user's individual reasoning style would require user-side contributions as the primary training signal—a design problem that remains open. Making this distinction explicit at the outset of a personal AI project determines which data collection strategy, segmentation method, and evaluation framework is appropriate, and prevents optimizing for the wrong artifact.

6 LIMITATIONS

We identify six limitations of the current work.

Single-user longitudinal design. This study analyzes one user's interaction history over three years, a design choice with explicit tradeoffs. The depth it enables (tracking structural properties across thousands of sessions, controlling for individual cognitive style across the full corpus, and isolating temporal ordering effects within a single user's reasoning patterns) is not achievable through multi-user aggregate approaches at comparable data resolution.

Single-user longitudinal design is not a limitation to be apologized for, but a methodological choice suited to personal AI research. Personal AI systems are designed to capture individual cognitive patterns, not population averages. A three-year interaction history from one user provides deeper temporal structure than cross-sectional snapshots from thousands of users.

We intentionally prioritize *internal validity* over *external validity*: the question this study addresses is whether trajectory structure exists at all in longitudinal interaction data and whether QA slicing removes it. Demonstrating the phenomenon's presence in a single corpus with high internal validity establishes feasibility and motivates multi-user replication as the natural next step. Attempting to claim cross-user generalizability from aggregate data would sacrifice the temporal depth required to observe structural properties in the first place. Direct generalizability claims across users are outside the scope of this study.

Whether the structural properties reported here—refinement chain length, transition dynamics, order dependence—are characteristic of personal interaction data broadly, or reflect this user's specific cognitive style, is an empirical question multi-user replication would address. We note that the methodological contributions of this work—the cognitive chunking pipeline, the graph schema, and the controlled comparison framework—are user-agnostic and designed for local replication: researchers can apply the same pipeline to their own interaction data without sharing raw conversation content, using the code and synthetic example data we release alongside this paper. We treat multi-user replication as the primary direction for future work rather than a limitation of the present design.

Rule-based segmentation. The cognitive chunking pipeline relies on lexical markers to detect cognitive event boundaries, capturing surface signals of reasoning transitions but potentially missing transitions without explicit linguistic markers. Incorporating embedding-based semantic distance as a segmentation criterion is a natural extension we leave for future work. Importantly, the main findings reported in Section 4—order dependence and non-recoverability—are measured on resulting graph structure and transition distributions, not derived from the segmentation rules themselves, and are therefore unlikely to be fully explained by this limitation. The shuffled-trajectory experiment (Group F) provides partial evidence that the observed structure is not an artifact of the segmentation rules: shuffling node order while preserving all segmentation decisions produces a significant distributional shift (KS $D = 0.101$, $p < 0.001$), indicating that structure depends on the temporal ordering of states, not on how those states were defined.

Cognitive nodes as text chunks. Graph nodes approximate cognitive states through rule-based text segmentation. A true cognitive state would require richer modeling: capturing epistemic status, confidence, and relation to prior beliefs, not just textual content. The gap between a text chunk and a cognitive state is a fundamental challenge for any approach that operationalizes cognition from interaction logs.

Consensus agent, not cognitive mirror. The source conversations are co-produced by a user and an assistant. Our graph representation captures textual transition structure in that dyadic history; it cannot identify which patterns belong to the user's individual cognition, the assistant's response policy, or their interaction. A cognitive mirror would require user-centered evidence and a different validation design.

Scope of findings. Our findings characterize one personal corpus under one segmentation method compared against one external QA baseline. The claim is not that cognitive trajectory graphs are the correct representation for personal AI training data, but that the choice of representation is consequential and has been underexamined. Alternative representations—hierarchical graphs, temporal knowledge bases, episodic memory structures—may capture different aspects of the same cognitive processes, and comparative evaluation across representation types is a productive direction for future work.

Curvature metric. Finally, we explored trajectory curvature as a second-order metric capturing the directional consistency of semantic transitions, but found that high-dimensional embedding spaces produce near-universal negative curvature values, making interpretation difficult. Developing curvature metrics robust to dimensionality effects is a direction for future work.

REFERENCES

[1] Jianlv Chen et al. 2024. BGE M3-Embedding: Multi-Lingual, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation. *arXiv preprint arXiv:2309.07597* (2024).

[2] Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, Dasha Metropolitansky, Robert Osazuwa Ness, and Jonathan Larson. 2024. From Local to Global: A Graph RAG Approach to Query-Focused Summarization. *arXiv preprint arXiv:2404.16130* (2024).

[3] Bent Flyvbjerg. 2006. Five Misunderstandings About Case-Study Research. *Qualitative Inquiry* 12, 2 (2006), 219–245.

[4] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-Rank Adaptation of Large Language Models. In *Proceedings of the International Conference on Learning Representations (ICLR '22)*.

[5] Evangelos Karapanos, Jhilmil Jain, and Marc Hassenzahl. 2012. Theories, Methods and Case Studies of Longitudinal HCI Research. In *Extended Abstracts of the SIGCHI Conference on Human Factors in Computing Systems (CHI EA '12)*. ACM, 2727–2730.

[6] Seoyeon Kim and Jaehyung Kim. 2026. SPRInG: Continual LLM Personalization via Selective Parametric Adaptation and Retrieval-Interpolated Generation. *arXiv preprint arXiv:2601.09974* (2026). arXiv:2601.09974.

[7] Nataliya Kosmyna, Eugene Hauptmann, Ye Tong Yuan, Jessica Situ, Xian-Hao Liao, Ashly Vivian Beresnitzky, Iris Braunstein, and Pattie Maes. 2025. Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task. *arXiv preprint arXiv:2506.08872* (2025).

[8] Xiang Lisa Li and Percy Liang. 2021. Prefix-Tuning: Optimizing Continuous Prompts for Generation. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics*. ACL, 4582–4597.

[9] Tao Long, Kendra Wannamaker, Jo Vermeulen, George Fitzmaurice, and Justin Matejka. 2025. Facilitating Longitudinal Interaction Studies of AI Systems. In *Adjunct Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (UIST '25 Adjunct)*. ACM.

[10] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. 2022. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, Vol. 35. Curran Associates, 27730–27744.

[11] Charles Packer, Vivian Fang, Shishir G. Patil, Kevin Lin, Sarah Wooders, and Joseph E. Gonzalez. 2023. MemGPT: Towards LLMs as Operating Systems. *arXiv preprint arXiv:2310.08560* (2023).

[12] Joon Sung Park, Joseph C. O'Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. 2023. Generative Agents: Interactive Simulacra of Human Behavior. In *UIST*.

[13] Pranav Rajpurkar, Jian Zhang, Konstantin Lopyrev, and Percy Liang. 2016. Squad: 100,000+ questions for machine comprehension of text. In *Proceedings of the 2016 conference on empirical methods in natural language processing*. 2383–2392.

[14] Lev Tankelevitch, Elena L. Glassman, Jessica He, Aniket Kittur, Mina Lee, Srishti Palani, Advait Sarkar, Gonzalo Ramos, Yvonne Rogers, and Hari Subramonyam. 2025. Understanding, Protecting, and Augmenting Human Cognition with Generative AI: A Synthesis of the CHI 2025 Tools for Thought Workshop. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*. ACM.

[15] Romal Thoppilan, Daniel De Freitas, Jamie Hall, Noam Shazeer, Apoorv Kulshreshtha, Heng-Tze Cheng, Alicia Jin, Taylor Bos, Leslie Baker, Yu Du, et al. 2022. LaMDA: Language Models for Dialog Applications. *arXiv preprint arXiv:2201.08239* (2022).

[16] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc V. Le, and Denny Zhou. 2022. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. In *Advances in Neural Information Processing Systems*, Vol. 35. Curran Associates, 24824–24837.

[17] Robert K. Yin. 2018. *Case Study Research and Applications: Design and Methods* (6th ed.). Sage.

[18] Saizheng Zhang, Emily Dinan, Jack Urbanek, Arthur Szlam, Douwe Kiela, and Jason Weston. 2018. Personalizing Dialogue Agents: I have a dog, do you have pets too?. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics*. ACL, 2204–2213.

[19] Sizhe Zhou. 2025. A Simple Yet Strong Baseline for Long-Term Conversational Memory of LLM Agents. *arXiv preprint arXiv:2511.17208* (2025).